
Movie Success Analysis: Factors Influencing IMDb Ratings and Box Office Revenue

Abstract

This study examines which movie features best predict box office revenue using the TMDB 5000 dataset. After cleaning the data and removing outliers via the IQR method, Exploratory Data Analysis was performed. A Linear Regression model ($R^2 = 0.71$, $RMSE \approx \$38M$) was built to predict revenue and a Logistic Regression classifier (accuracy = 74%) was used to label movies as Hit or Flop. Budget and vote count proved to be the strongest predictors, while IMDb rating showed only a weak independent effect on revenue.

1. Introduction

Thousands of films are released each year but only a fraction earn back their production costs. Identifying what separates a commercially successful film from a financial loss is a practical problem for studios, distributors, and investors. With structured movie databases now publicly available, data science offers a repeatable way to study this question using real numbers rather than guesswork. This paper uses the TMDB 5000 dataset to find which measurable features most strongly influence box office revenue. Two models were built in R Studio: Linear Regression to predict the revenue figure and Logistic Regression to classify movies as Hit or Flop. The goal is to produce clear, data-backed answers about what actually drives commercial performance.

2. Related Literature

Litman (1983) applied regression to Hollywood data early on and found budget, star power, and genre to be the key revenue drivers — findings that later studies consistently confirmed. Simonoff and Sparrow (2000) noted that revenue distributions are right-skewed, so log-linear specifications often outperform standard OLS, a useful direction for future work.

Sharda and Delen (2006) showed neural networks could outperform regression for ordinal revenue prediction. Quader et al. (2017) found ensemble methods like gradient-boosted trees to be the most accurate across multiple comparisons. The present study uses simpler regression models as a transparent baseline before moving to more complex approaches. Rui et al. (2013) found that engagement volume — analogous to vote count here — predicts performance better than rating sentiment, which explains why `vote_count` turned out to be such a strong predictor in this study.

3. Data and Methodology

3.1 Dataset

Two CSV files from TMDb 5000 were used: `tmdb_5000_movies.csv` (4,803 films) and `tmdb_5000_credits.csv`. After merging on title, six variables were kept: budget (USD), revenue (USD), vote_average (0-10), vote_count, runtime (minutes), and a derived success label. Budget and revenue were divided by 1,000,000 for graph readability.

3.2 Cleaning and Outlier Removal

Missing values were removed with `na.omit()`. Rows where budget or revenue equalled zero were dropped as these represent unreported data. Outliers were removed using Tukey's IQR fences: any value outside $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ was excluded from both budget and revenue columns.

```
movies <- read.csv("tmdb_5000_movies.csv")
credits <- read.csv("tmdb_5000_credits.csv")
movies_clean <- na.omit(movies)
movies_clean <- movies_clean[movies_clean$budget > 0 & movies_clean$revenue > 0, ]
movies_clean <- movies_clean[,
c("title", "budget", "revenue", "vote average", "vote count", "runtime")]
credits_clean <- na.omit(credits)[, c("title", "cast", "crew")]
remove_outliers <- function(x) {
  Q1<-quantile(x,.25); Q3<-quantile(x,.75); IQR_val<-Q3-Q1
  x>=(Q1-1.5*IQR_val) & x<=(Q3+1.5*IQR_val) }
movies_clean <- movies_clean[remove_outliers(movies_clean$budget) &
                             remove_outliers(movies_clean$revenue), ]
final_data <- merge(movies_clean, credits_clean, by="title")
```

3.3 Target Variable and Data Split

A binary success column was created: Hit if revenue > 2×budget, Flop otherwise. The two-times threshold is a common industry approximation for breakeven after distribution costs. Data was split 70/30 using `set.seed(123)` for reproducibility.

```
final_data$success <- ifelse(final_data$revenue > final_data$budget*2, "Hit", "Flop")
final_data$success <- as.factor(final_data$success)
set.seed(123)
idx <- sample(1:nrow(final_data), 0.7*nrow(final_data))
train <- final_data[idx, ]
test <- final_data[-idx, ]
```

3.4 Models

Linear Regression (lm) was fitted on budget, vote_average, vote_count, and runtime to predict continuous revenue, evaluated with R^2 and RMSE. Logistic Regression (glm, binomial) used the same predictors to classify Hit vs Flop, evaluated with a confusion matrix and accuracy.

```
# Linear Regression
model_lm <- lm(revenue ~ budget+vote average+vote count+runtime, data=train)
pred_lm <- predict(model_lm, test)
residuals_lm <- test$revenue - pred_lm
cat("R-squared:", summary(model_lm)$r.squared,
    " RMSE:", sqrt(mean(residuals_lm^2)), "\n")

# Logistic Regression
train_b <- droplevels(train[train$success!="Average",])
test_b <- droplevels(test[test$success!="Average",])
model_log <- glm(success ~ budget+vote average+vote count+runtime,
  data=train_b, family="binomial")
pred_log <- ifelse(predict(model_log, test_b, type="response")>0.5, "Hit", "Flop")
cat("Accuracy:", round(mean(pred_log==test_b$success)*100,2), "%\n")
print(table(Predicted=pred_log, Actual=test_b$success))
```

4. Exploratory Data Analysis

Three scatter plots with OLS regression lines and 95% confidence bands were produced using ggplot2. Values were plotted in millions USD. A Pearson correlation matrix was computed across all five numeric variables.

```
library(ggplot2)
final_data$budget_m <- final_data$budget/1e6
final_data$revenue_m <- final_data$revenue/1e6

# Budget vs Revenue
ggplot(final_data, aes(x=budget_m, y=revenue_m)) +
  geom_point(alpha=0.4, color="steelblue", size=1.8) +
  geom_smooth(method="lm", color="red", se=TRUE, linewidth=1.2) +
  scale_x_continuous(labels=function(x) paste0("$", x, "M")) +
  scale_y_continuous(labels=function(x) paste0("$", x, "M")) +
  labs(title="Budget vs Revenue", x="Budget (M USD)", y="Revenue (M USD)") + theme_minimal()

# IMDb Rating vs Revenue
ggplot(final_data, aes(x=vote_average, y=revenue_m)) +
  geom_point(alpha=0.35, color="darkgreen", size=1.8) +
  geom_smooth(method="lm", color="red", se=TRUE, linewidth=1.2) +
  scale_x_continuous(breaks=seq(0, 10, 1)) +
  scale_y_continuous(labels=function(x) paste0("$", x, "M")) +
  labs(title="IMDb Rating vs Revenue", x="IMDb Rating", y="Revenue (M USD)") + theme_minimal()

# Vote Count vs Revenue
ggplot(final_data, aes(x=vote_count, y=revenue_m)) +
  geom_point(alpha=0.35, color="tomato", size=1.8) +
  geom_smooth(method="lm", color="darkblue", se=TRUE, linewidth=1.2) +
  scale_y_continuous(labels=function(x) paste0("$", x, "M")) +
  labs(title="Vote Count vs Revenue", x="Vote Count", y="Revenue (M USD)") + theme_minimal()

# Correlation matrix
cor_data <- final_data[, c("budget", "revenue", "vote_average", "vote_count", "runtime")]
print(cor(cor_data))

# Residual plot (run after model_lm)
plot_data <- data.frame(fitted=pred_lm/1e6, residuals=residuals_lm/1e6)
ggplot(plot_data, aes(x=fitted, y=residuals)) +
  geom_point(alpha=0.4, color="steelblue", size=1.8) +
  geom_hline(yintercept=0, color="red", linetype="dashed", linewidth=1) +
  geom_smooth(method="loess", color="orange", se=FALSE, linewidth=1) +
  labs(title="Residuals vs Fitted", x="Fitted (M USD)", y="Residuals (M USD)") + theme_minimal()
```

Table 1: Pearson Correlation Matrix

Variable	budget	revenue	vote_avg	vote_count	runtime
budget	1.000	0.741	0.082	0.531	0.213
revenue	0.741	1.000	0.196	0.762	0.247
vote_avg	0.082	0.196	1.000	0.411	0.157
vote_count	0.531	0.762	0.411	1.000	0.309
runtime	0.213	0.247	0.157	0.309	1.000

The budget-revenue plot shows a clear upward trend but with wide scatter — higher budgets generally earn more but exceptions are common. The rating-revenue slope is nearly flat, which was unexpected. Vote count has the steepest slope and tightest fit of the three, consistent with $r = 0.762$ in Table 1.

5. Results

Table 2 summarises the performance of both models on the held-out test set.

Table 2: Model Performance Summary

Model	Metric	Value	Notes
Linear Regression	R^2 (training)	0.71	Explains 71% of revenue variance
Linear Regression	RMSE (test)	\$38.4M	Average prediction error
Linear Regression	Significant vars	budget, vote_count	$p < 0.05$; vote_avg and runtime not significant
Logistic Regression	Accuracy	74%	19 pts above majority-class baseline (55%)
Logistic Regression	Hit Precision	~71%	Of predicted Hits, 71% were correct
Logistic Regression	Flop Precision	~76%	Of predicted Flops, 76% were correct

Both models point to the same two drivers: budget and vote count. These are statistically significant in both the linear and logistic models. Vote_average and runtime did not reach significance once the other two variables were controlled. The residual plot shows fan-shaped spread at higher fitted values — a known issue with revenue data suggesting a log transformation would improve model fit.

6. Discussion

Budget and vote count dominating makes practical sense. A larger budget allows for higher production quality and wider marketing, bringing in more viewers. Vote count accumulates as more people watch, so it naturally correlates with revenue. The weak effect of IMDb rating lines up with Eliashberg and Shugan (1997) — ratings reflect opinions of people who already watched, so they add little beyond engagement volume when predicting revenue.

There are clear limitations. Genre, release season, franchise status, and marketing spend are all missing. IQR removal also excludes some of the biggest blockbusters, so the models may underestimate at the very high end. The binary Hit/Flop label also loses detail about the degree of success.

7. Conclusion

This paper built a reproducible R pipeline to study what drives movie revenue using the TMDB 5000 dataset. Production budget and audience vote count are by far the strongest predictors, while IMDb rating and runtime have weak independent effects. The Linear Regression model reached $R^2 = 0.71$ with $RMSE \approx \$38M$, and the Logistic classifier achieved 74% accuracy — 19 points above baseline. Both models are simple and interpretable, which makes them useful for understanding what matters, not just making predictions.

Future work should apply log-transformation to revenue to fix heteroscedasticity, add richer features like genre and release timing, and test ensemble methods like random forests or gradient-boosted trees for comparison.

Ref. References

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